



Morning Brief

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A personal learning and awareness brief on AI governance, cybersecurity, the federal landscape, and the science behind the methods. Every news item below is sourced; analysis is labeled as such.

AI Governance & Policy

- **Executive Order 14409, "Promoting Advanced Artificial Intelligence Innovation and Security" (June 2, 2026).** It directs upgrades to federal cyber defenses, a *voluntary* framework for frontier-model developers to engage with the government before deployment, and enforcement against criminal misuse of AI — explicitly **without** a mandatory licensing or preclearance regime. Timelines are aggressive: 30 days for cyber directives, 60 for the frontier framework. [White House](#) · [Covington analysis](#) — *Why it matters*: the voluntary, NIST-aligned posture mirrors the AI Framework's own design and is the federal counterpart to the agentic/groundedness overlays in the v3.0 roadmap.
- **EU AI Act enforcement against general-purpose AI providers begins August 2, 2026.** From that date the European Commission can supervise and fine GPAI model

providers (up to €15M or 3% of global revenue), ending the one-year grace period that opened August 2, 2025. The GPAI Code of Practice covers Transparency, Copyright, and Safety & Security. [EU AI Act — Enforcement of Chapter V](#) — *Why it matters*: the binding international benchmark for GenAI obligations; useful calibration for any transparency/groundedness requirement you scope.

- **Colorado rewrote the most comprehensive U.S. state AI law before it took effect.** The 2024 Colorado AI Act (SB 24-205) was repealed and replaced by SB 26-189 (Automated Decision-Making Technology), signed May 14, 2026 — and the framework-based safe harbor did **not** survive into the new statute. [Goodwin — Congress and State Lawmakers Race to Keep Up With AI](#) — *Why it matters*: the disappearing safe harbor signals that "adopt a recognized framework" may not, by itself, be a compliance shield — relevant to how you position the framework's assurance value.
- **NIST's agentic-AI standards work continues to mature** (AI 100-5 agentic profile; an agent-identity/authentication emphasis; RFI under docket NIST-2025-0035), even as FedScoop reports agentic AI is reaching agencies faster than the guardrails. [FedScoop — "Agentic AI is coming to government faster than its guardrails"](#) — *Why it matters*: this is the independent triangulation your v3.0 agentic-overlay note asks for — evidence beyond OWASP's Agentic Top 10 before any control is drafted.
- **NIST is also building an AI RMF Profile for Trustworthy AI in Critical Infrastructure** (concept note released April 7, 2026), with SP 800-53 Control Overlays for AI and RMF addenda expected later in 2026, following the Cybersecurity Framework Profile for AI (NIST IR 8596, December 2025). [NIST — AI Risk Management Framework](#) — *Why it matters*: a sector-profile pattern you can borrow for an IC-specific overlay.

Cybersecurity & Threats

- **Microsoft's June 2026 Patch Tuesday was its largest on record** — roughly 200 vulnerabilities fixed, including multiple publicly disclosed zero-days: a BitLocker security-feature bypass (CVE-2026-50507), an HTTP.sys denial-of-service (CVE-2026-49160), and a Windows CTFMON elevation-of-privilege to SYSTEM (CVE-2026-45586). [BleepingComputer](#) ·

[Zero Day Initiative review](#) — *Federal relevance*: the volume alone strains agency patch cycles already compressed by EO 14409.

- **CISA added two actively exploited vulnerabilities on June 15**: CVE-2026-20262 (Cisco Catalyst SD-WAN Manager path traversal) and CVE-2026-54420 (LiteSpeed cPanel plugin symlink-following → root). FCEB remediation is due **June 18 — tomorrow**. [CISA alert](#) · [Security Affairs](#)
- **CISA added three more on June 9**: an Arista EOS flaw, a Google Chromium V8 out-of-bounds read/write, and a separate Cisco Catalyst SD-WAN Manager output-encoding issue. [CISA alert](#) — *Pattern*: two Cisco SD-WAN Manager entries in one week point straight at agency management planes.
- **Check Point's June 8 threat report flags a critical Cisco Unified Communications Manager flaw** (unauthenticated file write → root) alongside continued AI-assisted malware tradecraft. [Check Point Research](#)
- **The ShinyHunters extortion wave widened in mid-June**, with the group claiming ~297GB from the Council of Europe and leaking data tied to U.S. dental-benefits administrator DentaQuest (millions of accounts, including government IDs and health-insurance details). [TechRepublic](#) · [TechCrunch](#) — [worst breaches of 2026 so far](#) — *Why it matters*: identity + health data theft at scale keeps third-party/supply-chain exposure at the top of the risk picture.
- **EO 14409 sets up new cyber directives and an AI cybersecurity clearinghouse**. CISA must issue Binding Operational Directives within 30 days; Treasury, NSA, and CISA are to stand up a voluntary clearinghouse to coordinate vulnerability scanning, validation, and patch prioritization with industry and critical-infrastructure operators. [Federal News Network](#) — *Why it matters*: compresses the federal patch cycle and raises the payoff of automated KEV-to-asset matching.

On the Federal Front

- **SASC completed markup of the FY2027 NDAA (June 8–10) and advanced it 18–9**. The draft creates an **Undersecretary of Defense for Cyber, Information, and Networks** — dual-hatted as DoD CIO and principal cyber advisor — with the CDAO nested inside that office. [DefenseScoop](#) · [SASC press release](#)

— *Why it matters*: if enacted, it consolidates where AI and data-governance authority sit in the Department.

- **GenAI.mil — the DoW's enterprise generative-AI platform — is scaling fast**, with the CDAO reporting 100,000 agents built within two weeks of launch and plans to put commercial models in front of essentially all military and civilian personnel. [DefenseScoop — Pentagon uses GenAI.mil to create 100K agents](#) · [GovCIO](#) — *Why it matters*: mass adoption with that velocity is exactly where T&E, groundedness, and provenance controls must keep pace.
- **DIU's Thunderforge pushes commercial AI into operational and theater-level planning**, fielding first with INDOPACOM and EUCOM to speed course-of-action generation and AI-assisted wargaming. [DIU](#) — *Why it matters*: decision-support at the planning tier raises the human-oversight (C2) questions your framework scores.
- **Congress is moving DOD-AI guardrails in parallel with the NDAA**. Rep. Adam Schiff's HALO Act (Human Authority in Lethal Operations) would expand DoD AI safety/ethics requirements, mandate impact assessments for high-risk capabilities, and require rigorous testing before fielding AI-enabled weaponry; other bills target AI's labor and data-center-energy impacts. [Nextgov/FCW](#) · [Akin — AI measures in defense legislation](#)
- **EO 14409 implementation lands on agencies first**: CISA BODs within 30 days, the Treasury/NSA/CISA clearinghouse, and accelerated hiring of cybersecurity specialists within 60 days. [White House](#) — *Why it matters*: concrete federal cyber and AI-engagement mechanics could appear by July.
- **CDAO is extending AI into contested-logistics problems**, per officials at a June 3 industry panel — taking the office's enterprise GenAI and "Wingman" digital-assistant work into operational sustainment. [DefenseScoop](#)

AI Industry Landscape

- **The coding-model war went four-way**. On June 1, Microsoft and Google launched rival AI coding models, taking direct aim at Anthropic's Claude and OpenAI's franchise in the most lucrative enterprise-AI use case. [CNBC](#) — *Why it matters*: coding is

where model quality converts to revenue fastest; expect aggressive price/perf moves.

- **The EU's "Cloud and AI Development Act" (announced June 3)** would bar AWS, Microsoft Azure, and Google Cloud from the most sensitive EU government workloads over U.S. CLOUD Act exposure, and require a sovereignty risk assessment before contracting for critical-sector cloud; the EU Parliament also swapped Google for Qwant as its default search June 4. [The Times tech report](#) — *Why it matters*: data sovereignty is now a procurement weapon reshaping where U.S. hyperscalers can sell.
 - **Anthropic and OpenAI both pushed into enterprise services, not just models.** Anthropic stood up an enterprise-AI joint venture with institutional partners, and OpenAI launched a consulting/services arm — a new front in the enterprise race. [TechCrunch](#) · [CIO](#) — *Why it matters*: labs moving into delivery/integration changes how buyers (including government SIs) procure.
 - **Anthropic confirmed a confidential IPO filing and reported run-rate revenue past \$30B** (up from roughly \$9B at end-2025), while deepening multi-gigawatt compute via Google and Broadcom TPUs. [CIO](#) · [Anthropic newsroom](#) — *Why it matters*: a public-market path plus locked-in compute signals durability of the frontier-lab tier.
 - **The OpenAI–Microsoft partnership was restructured**, freeing OpenAI to run on AWS, Google, and other clouds beyond Azure — loosening the tightest lab-cloud coupling in the market. [The New Stack](#) — *Why it matters*: multi-cloud frontier models reshuffle the hyperscaler competitive map.
 - **In the national-security lane, the field is set without Anthropic.** Since the Pentagon's May 1 clearance of eight firms (SpaceX, OpenAI, Google, NVIDIA, Reflection, Microsoft, AWS, Oracle) for IL6/IL7 networks, Anthropic stayed out over weapons/surveillance guardrails and Palantir pulled Claude from DoD platforms; meanwhile GenAI.mil is reportedly bringing xAI's Grok to IL5. [Breaking Defense](#) · [DefenseScoop](#) — *Why it matters*: vendor red lines are now a market-structuring force in defense AI.
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Project Status

Most recently active first; all are live.

- **NetMon-Server** (June 15) — single-file, loopback-only Windows network-monitor and light SIEM with an optional on-device (Ollama) AI layer scoring posture across six NIST-aligned axes. *Next action:* given today's June 18 KEV deadline, add a rule matching the local listening-port/process inventory against the live KEV catalog (Cisco SD-WAN Manager, LiteSpeed cPanel) and surface a posture hit.
- **MongoDB** (June 12) — vetting-system research tracker with DFD and system-flow diagrams. *Next action:* consolidate the duplicate ... - Copy diagram sources into canonical .mermaid/.txt files and add a short README so the artifact set is self-describing.
- **AI Framework** (June 10) — the governance framework; the v3.0 roadmap defers an agentic-AI overlay and a GenAI groundedness overlay, each flagged as single-source. *Next action, tied to today's news:* open the agentic-overlay evidence re-verification using NIST's AI 100-5 agentic profile, EO 14409's frontier-framework language, and the FedScoop "guardrails lagging" reporting as triangulation beyond OWASP.
- **AI Research vs. Failures** (active) — a dual-track pipeline that ingests AI/ML research papers and documented AI incidents, applies NLP topic modeling and supervised classification, and surfaces the gap between what is researched and where real-world harm occurs, plus a quantum-ML governance gap analysis. *Next action:* refresh the incident corpus and re-run the research-vs-harm gap classification, treating this week's EO 14409 and NIST agentic-AI items as new policy-attention signal to test against the incident clusters.
- **Bathymetry (remote sensing)** (active) — a raster-to-graph GNN pipeline that estimates seafloor depth from geospatial rasters, staged across notebooks from environment setup and data acquisition through raster-to-graph conversion, GNN training, and scaling. *Next action:* finish the raster-to-graph step (03) and validate GNN training (04) against a held-out set of depth soundings before scaling up.
- **Quantum-Inspired Machine Learning Models** (planned) — implement variational quantum circuits and quantum kernels, integrate them into hybrid (quantum-classical) pipelines, and

benchmark against classical models on small-data and structured-data tasks. Not yet scaffolded. *Next action*: stand up a minimal PennyLane (or Qiskit) environment and reproduce one variational-classifier and one quantum-kernel baseline on a small structured dataset, with a classical model (SVM/gradient boosting) as the head-to-head control.

- **Geospatial Imaging — Open-Water Analysis** (active) — a geospatial imaging project focused on open-water analysis from remote-sensing imagery. *Next action*: scaffold the data-acquisition pipeline (imagery source plus `.env` per the project's secrets policy) and define the open-water detection target and a quantitative success metric.

Five Project Briefs

1. Tensor-network (DMRG/MPS) ground states of quantum spin chains

- **Domain**: Condensed-matter / quantum many-body physics · **Language(s)**: Julia (ITensor.jl) with a Python (quimb) cross-check.
- **Problem statement**: The dimension of a quantum many-body Hilbert space grows as 2^N , so exact diagonalization stalls around 18–20 spins on a laptop. Yet most physically relevant ground states of one-dimensional, gapped, local Hamiltonians live in a tiny, low-entanglement corner of that space. Matrix product states (MPS) exploit exactly that structure, and the density-matrix renormalization group (DMRG) finds them efficiently. Building intuition here is the gateway to modern computational quantum physics and to tensor-network methods now crossing into machine learning.
- **Objective(s)**: Implement and validate a DMRG solver for the spin-1/2 Heisenberg and transverse-field Ising chains; recover ground-state energies and correlation functions; and map a quantum phase transition by sweeping a control parameter.
- **Questions to answer**:
 - How does ground-state energy error scale with MPS bond dimension, and where is the accuracy/cost knee?
 - Can entanglement entropy versus subsystem size distinguish the gapped from the critical phase of the transverse-field Ising chain?

- Where does the quantum critical point sit, and does finite-size scaling of the order parameter locate it correctly?
- **Proposed method/approach:** Construct the Hamiltonian as a matrix product operator; run two-site DMRG sweeps with increasing bond dimension and a truncation-error stopping rule; extract energy, magnetization, and two-point correlators; compute von Neumann entanglement entropy across a cut; sweep the transverse field through the critical point and apply finite-size scaling; cross-check a small chain against exact diagonalization and quimb.
- **Tools:** Julia + ITensor.jl (primary); Python + quimb and NumPy (cross-check); matplotlib/Makie. No external dataset — states are simulated. Laptop CPU; no GPU required.
- **Scope:** In scope — 1-D nearest-neighbor chains up to ~100 sites, open boundaries, two models, one phase transition. Out of scope — 2-D systems, real-time evolution (TEBD), finite-temperature methods. Deliverables — a reproducible Julia notebook, an energy-vs-bond-dimension convergence study, and an annotated phase-transition figure.
- **Estimated duration to complete:** 2–3 weeks part-time.
- **Sources:** [ITensor docs](#); [Schollwöck \(2011\), Ann. Phys. 326](#).

2. Few-shot bioacoustic species ID from frozen audio foundation-model embeddings

- **Domain:** Ecology / bioacoustics (passive acoustic monitoring) ·
- **Language(s):** Python (PyTorch).
- **Problem statement:** Passive acoustic monitoring generates enormous audio archives, but labeled examples for any given species — especially rare or regional ones — are scarce, and training a deep classifier per deployment is infeasible. Frozen audio foundation models (Perch / BirdNET-class embeddings) turn raw spectrogram audio into general-purpose vectors, after which a lightweight few-shot classifier can recognize new species from a handful of clips. This is the dominant emerging pattern for low-label ecological monitoring.
- **Objective(s):** Build a pipeline that embeds field recordings with a frozen pretrained model and classifies species with a prototypical / nearest-centroid head from as few as 5–20 labeled clips per class, with calibrated confidence.
- **Questions to answer:**

- How does accuracy scale with labeled clips per species (1, 5, 10, 20-shot)?
- Do frozen foundation-model embeddings beat hand-engineered mel-feature baselines under the same label budget?
- How much does recorder/domain shift and background noise degrade few-shot transfer?
- **Proposed method/approach:** Pull a labeled subset from a public corpus (BirdCLEF / Xeno-canto); slice into fixed windows and generate frozen-model embeddings; build class prototypes as embedding centroids and classify by cosine distance; compare against a small logistic-regression head and a mel-spectrogram CNN trained on the same few shots; add temperature scaling and report reliability diagrams; stress-test with additive field noise.
- **Tools:** Python, PyTorch, torchaudio, a frozen Perch/BirdNET embedding model, scikit-learn; datasets — BirdCLEF (Kaggle), Xeno-canto. A single consumer GPU accelerates embedding extraction; cached embeddings train on CPU.
- **Scope:** In scope — 10–25 species, few-shot heads, calibration, one noise study. Out of scope — full backbone fine-tuning, overlapping-call separation, real-time on-device inference. Deliverables — an embedding cache, a few-shot benchmark table, calibration plots, a short methods write-up.
- **Estimated duration to complete:** 3–4 weeks part-time.
- **Sources:** [Ghani et al., Sci. Rep. \(2023\)](#); [BirdCLEF on Kaggle](#).

3. Non-stationary extreme-value analysis of climate extremes

- **Domain:** Climate science / statistical hydrology · **Language(s):** R (extRemes, ismev), Python (pyextremes) cross-check.
- **Problem statement:** Infrastructure design and risk pricing rely on return levels — the "1-in-100-year" heat or rainfall event. Classical extreme-value theory assumes a stationary climate, but warming makes the distribution of extremes drift. Fitting non-stationary GEV and peaks-over-threshold (GPD) models with time- or covariate-dependent parameters is the statistically correct way to estimate how return levels are changing — and an R strength.
- **Objective(s):** Fit stationary and non-stationary GEV/GPD models to a long station record, test whether non-stationarity is

warranted, and produce return-level curves with uncertainty for past versus present climate.

- **Questions to answer:**

- Is a trend in the GEV location (and/or scale) supported over a constant-parameter model by likelihood-ratio test and AIC?
- How much has the 50- and 100-year return level shifted between the early and late record?
- Are profile-likelihood intervals materially wider than delta-method intervals at long return periods?

- **Proposed method/approach:** Pull a multi-decade daily series (GHCN-Daily) or an ERA5 grid cell; form annual block maxima and a peaks-over-threshold series with mean-residual-life threshold selection; fit stationary GEV/GPD, then add a linear time covariate to location/scale; compare via likelihood-ratio test and AIC; compute return levels and profile-likelihood intervals; diagnose with QQ and return-level plots; cross-check one fit in `pyextremes`.

- **Tools:** R with `extRemes`, `ismev`, `evd`; data — NOAA GHCN-Daily or Copernicus ERA5; Python `pyextremes`. Laptop CPU.

- **Scope:** In scope — one to three stations/cells, GEV and GPD, one covariate, full diagnostics. Out of scope — spatial pooling, Bayesian hierarchical EVT, compound/multivariate extremes. Deliverables — model-comparison table, return-level figures with intervals, interpretive note.

- **Estimated duration to complete:** 2–3 weeks part-time.

- **Sources:** [Coles \(2001\)](#), [Springer](#); `extRemes`.

4. Adjoint-method differentiable inverse design of nanophotonic structures

- **Domain:** Photonics / computational electromagnetics ·

- **Language(s):** Julia (or Python with `autodiff`).

- **Problem statement:** Designing a photonic device — a wavelength splitter, a mode converter — has historically meant expert intuition plus parameter sweeps. Inverse design flips this: define a figure of merit, then use the adjoint method to get the gradient of that objective with respect to every pixel of the permittivity distribution in a single extra simulation, and optimize directly. This "topology optimization for light" is how modern compact photonic components are designed, and small 2-D problems are laptop-feasible.

- **Objective(s):** Implement a 2-D frequency-domain electromagnetic solver, derive/verify the adjoint gradient, and inverse-design a compact waveguide component that meets a transmission target.
- **Questions to answer:**
 - Does the adjoint gradient match a finite-difference gradient to acceptable tolerance (the correctness gate)?
 - Can gradient-based optimization reach a target transmission/splitting ratio a naive sweep cannot within the same budget?
 - How do fabrication-aware constraints (minimum feature size, binarization) trade off against performance?
- **Proposed method/approach:** Build a 2-D finite-difference frequency-domain (FDFD) Maxwell solver on a modest grid; define an output-port transmission figure of merit; implement the adjoint solve and validate the gradient against finite differences; optimize the permittivity field with projected gradient / L-BFGS; add a density filter and projection for minimum feature size and grayscale-to-binary control; report optimized layout and field maps.
- **Tools:** Julia (custom FDFD + sparse solver) or Python (autograd/JAX, or the `ceviche` differentiable FDFD); matplotlib/Makie. CPU sufficient for 2-D; no external dataset.
- **Scope:** In scope — one 2-D component, single frequency, one fabrication constraint, gradient validation. Out of scope — 3-D FDTD, broadband multi-objective design, tolerance/yield analysis. Deliverables — a validated adjoint-gradient notebook, an optimized device layout, before/after field plots.
- **Estimated duration to complete:** 4–5 weeks part-time.
- **Sources:** [Molesky et al., Nature Photonics 12 \(2018\)](#); `ceviche`.

5. DeepONet operator surrogate for lithium-ion battery state-of-health forecasting

- **Domain:** Energy storage / materials informatics · **Language(s):** Python (PyTorch).
- **Problem statement:** Battery degradation depends on the entire history of how a cell was cycled — a mapping from an input *function* (the current/temperature profile) to an output *function* (capacity fade over cycles). Standard regressors treat fixed feature vectors; operator-learning networks such as DeepONet learn maps between function spaces directly, a better-matched

and increasingly popular inductive bias for path-dependent physical systems. A laptop and public cycling datasets are enough to test whether it beats conventional baselines.

- **Objective(s):** Train a DeepONet to predict a cell's state-of-health trajectory from its usage profile, and benchmark it against feature-based gradient boosting and an LSTM.
- **Questions to answer:**
 - Does an operator-learning surrogate generalize to unseen charging profiles better than a feature-based or recurrent baseline?
 - How far ahead (cycles) can remaining-useful-life be forecast before error exceeds a usable threshold?
 - Which input channels (current, voltage, temperature) carry the most predictive signal for capacity fade?
- **Proposed method/approach:** Load a public cycling dataset (NASA Prognostics or the Severson/Toyota fast-charging set); frame inputs as sampled usage functions and outputs as capacity-vs-cycle curves; build a DeepONet (branch net over the profile, trunk net over the query cycle); train with held-out cells and compare to XGBoost on engineered features and to an LSTM; evaluate with RMSE, remaining-useful-life error, and deep-ensemble prediction intervals; ablate input channels.
- **Tools:** Python, PyTorch (or DeepXDE), scikit-learn, XGBoost; datasets — NASA Battery Prognostics or Severson et al. A single consumer GPU trains in minutes to an hour; CPU works for small runs.
- **Scope:** In scope — one chemistry/dataset, three model families, an uncertainty estimate, one ablation. Out of scope — physics-based DFN modeling, cross-chemistry transfer, on-BMS deployment. Deliverables — a trained surrogate, a cross-family benchmark table, RUL forecast plots with intervals.
- **Estimated duration to complete:** 3–4 weeks part-time.
- **Sources:** [Lu et al., Nat. Mach. Intell. 3 \(2021\)](#); [Severson et al., Nat. Energy 4 \(2019\)](#).

Training

1. Python — Writing your own decorators with `functools.wraps`

A decorator is a function that wraps another function to add behavior — logging, timing, retries, access control — without editing the

wrapped function's body. Understanding how to *write* one (not just apply `@something`) is what lets you factor cross-cutting concerns out of business logic cleanly.

```
import functools
import time

def timed(func):
    # Wrap a function so each call prints how long it took
    @functools.wraps(func) # Preserve func's name, docstring, etc.
    def wrapper(*args, **kwargs):
        start = time.perf_counter() # High-resolution timer
        result = func(*args, **kwargs) # Call the original function
        elapsed = time.perf_counter() - start
        print(f"{func.__name__} took {elapsed:.6f}s") # Print timing
        return result # Always return what the wrapped function returns
    return wrapper

@timed
def slow_square(n):
    """Return n squared after a brief pause."""
    time.sleep(0.05)
    return n * n

print(slow_square(9)) # Prints timing line, then 81.
print(slow_square.__name__) # 'slow_square', not 'wrapper'
```

Why it matters: decorators keep the wrapped function readable and make the added behavior reusable across many functions. **Common pitfall:** omitting `@functools.wraps(func)`. Without it, the wrapper clobbers the original function's `__name__`, `__doc__`, and signature, which breaks introspection, documentation tools, and some frameworks that dispatch on function metadata.

Reference: [Python docs — functools.wraps](#).

2. R — Tidying model output with broom

Base R model objects (`lm`, `glm`, `nls`, `htest`) print nicely but are awkward to program against — coefficients live in one structure, fit statistics in another. The `broom` package converts any model into tidy data frames with three verbs: `tidy()` for per-term estimates, `glance()` for one-row model-level statistics, and `augment()` for per-observation fitted values and residuals.

```
library(broom)
library(dplyr)

# Fit a simple model on the built-in mtcars data.
model <- lm(mpg ~ wt + hp, data = mtcars)

# Per-term estimates: term, estimate, std.error, statistic, p.value
tidy(model, conf.int = TRUE)

# One-row model summary: r.squared, adj.r.squared, AIC, BIC, etc.
glance(model)

# Original data plus .fitted, .resid, .hat, .cooks, etc.
augment(model) %>%
  arrange(desc(abs(.resid))) %>% # Largest residuals first
  head(3)
```

Because the output is a data frame, you can `bind_rows()` results across many models, feed `tidy()` straight into `ggplot2` for coefficient plots, or filter terms by p-value programmatically. This pairs naturally with list-columns: fit a model per group with `purrr::map()`, then `map_dfr(models, tidy)` to stack every group's coefficients into one tidy table.

Reference: [broom documentation](#).

3. Data-centric — Schema evolution and the schema registry

When producers and consumers of data evolve independently, the schema *will* change — a new field, a renamed column, a widened type. A **schema registry** (used with Avro, Protobuf, or JSON Schema in streaming systems like Kafka) stores versioned schemas and enforces a **compatibility mode** so a change can't silently break downstream readers.

The compatibility modes are the concept worth internalizing:

- **Backward compatible** — new schema can read data written with the old schema. Safe to *add an optional field with a default* or *remove a field*. Lets you upgrade **consumers first**.
- **Forward compatible** — old schema can read data written with the new schema. Lets you upgrade **producers first**.
- **Full compatible** — both directions hold. The safest, most constrained option.

A practical rule for Avro: give every new field a default, and never reuse or repurpose a field name. A field with a default is removable

under backward compatibility and skippable under forward compatibility, which is what keeps a rolling deployment from corrupting reads mid-migration.

```
{
  "type": "record",
  "name": "Event",
  "fields": [
    { "name": "id", "type": "string" },
    { "name": "ts", "type": "long" },
    { "name": "source", "type": ["null", "string"], "default": null }
  ]
}
```

Here `source` is nullable with a default, so adding it is a backward-compatible change: old readers ignore it, new readers get `null` for historical records. Reference: [Confluent — Schema Evolution and Compatibility](#).

4. AI method — Hamiltonian Monte Carlo (HMC)

What it is. HMC is a Markov chain Monte Carlo sampler for drawing from a probability distribution $p(\theta)$ known only up to a constant — the everyday situation in Bayesian inference, where $p(\theta \mid \text{data})$ is the unnormalized posterior. Where random-walk Metropolis stumbles around parameter space and mixes slowly in high dimensions, HMC proposes distant, high-acceptance moves by simulating physics.

Intuition / math. Treat the negative log-posterior as a potential energy $U(\theta) = -\log p(\theta)$ and introduce an auxiliary momentum r with kinetic energy $K(r) = r^T M^{-1} r / 2$. The total energy is the Hamiltonian $H(\theta, r) = U(\theta) + K(r)$. To propose a new state, sample a fresh momentum, then integrate Hamilton's equations with the **leapfrog** integrator for L steps of size ϵ . Hamiltonian dynamics conserves energy and volume, so a proposal that follows the gradient of the log-posterior lands in a region of comparable probability — far from the start yet rarely rejected. A Metropolis correction at the end accounts for the integrator's small discretization error, keeping the chain exact.

When to use it. Reach for HMC on continuous, differentiable posteriors of moderate to high dimension — hierarchical regressions, latent-variable models — where you can compute $\nabla \log p(\theta)$. It needs gradients (so it does not apply to discrete parameters directly) and is sensitive to ϵ and L ; the No-U-Turn Sampler (NUTS) in Stan, PyMC, and NumPyro tunes both automatically, which is why HMC/NUTS is the default engine in modern probabilistic programming. Today's

companion notebook implements leapfrog HMC from scratch and tests it against a known target.

Reference: [Betancourt, "A Conceptual Introduction to Hamiltonian Monte Carlo" \(2017\).](#)

See notebooks/2026-06-17-hamiltonian-monte-carlo.ipynb for a runnable from-scratch implementation with explicit Modeling & Simulation and Test & Evaluation sections.

Signal & Trajectory

Analysis and inference — not reported fact. Each point traces to an item above.

- If EO 14409's 30/60-day clocks hold (Governance; Federal Front), expect the first CISA Binding Operational Directives, the Treasury/NSA/CISA AI cybersecurity clearinghouse, and the frontier- engagement framework draft by mid-to-late July; watch whether "voluntary" survives contact with the parallel criminal-misuse enforcement push and the HALO Act's mandatory-testing language.
- If CISA keeps adding Cisco Catalyst SD-WAN Manager flaws (Cybersecurity — two in one week) on top of a record ~200-fix Patch Tuesday, expect management-plane appliances and patch-cycle capacity to be the near-term federal squeeze; watch the June 18 KEV deadline as the immediate test, and treat automated KEV-to-asset matching (the NetMon-Server next action) as the practical hedge.
- If GenAI.mil keeps scaling at 100K-agents-in-two-weeks velocity and Thunderforge fields with INDOPACOM/EUCOM (Federal Front), expect adoption to outrun T&E, groundedness, and human-oversight controls — exactly the gap your v3.0 overlays target; watch the FY27 NDAA Undersecretary language for where that governance authority lands.
- If the EU's Cloud and AI Development Act and the AI Act's August 2 GPAI enforcement land in the same window (Industry; Governance), expect U.S. hyperscalers and labs to face a two-front European squeeze; watch for sovereign-cloud joint ventures and the OpenAI–Microsoft multi-cloud unwind as the likely responses.
- If the four-way coding-model fight (Industry) compresses price and quality, expect faster enterprise adoption but sharper lock-in

questions; watch whether Anthropic's services JV and IPO path, set against its national-security exclusion, widen the gap between its commercial and government trajectories.

Ledgers updated: `project-ideas/_ledger.md`, `training-ledger.md`, and `publication-ledger.md` were appended for this edition (No. 7).

Attached: brief (.pdf) + companion notebook (.ipynb) + project briefs (.docx)

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